Power BI Lab Practicals 7 - 9

RetailPulse Sales Analysis

Business Analytics / Descriptive Analytics Course | IBM Cognos Analytics & Power BI Lab Manual

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Practicals:	7 (Import & Transform), 8 (Model, DAX & Visuals), 9 (Time Intelligence & Advanced DAX)
Dataset:	RetailPulse (Sales, Product, Store tables)
Date:	09 October 2026

Dataset Used: RetailPulse (clean)

A clean synthetic RetailPulse workbook, **RetailPulse_Sales.xlsx**, was used with three tables: **Sales** (1,800 rows, Jan 2024 - Dec 2025), **Product** (12 products, 4 categories) and **Store** (8 stores, 4 regions). Two full calendar years are present so the year-on-year measures in Practical 9 return values.

Sales table (first 6 rows)

OrderID	Sales Date	ProductID	StoreID	Quantity	Sales Amount
ORD100001	2024-01-01	P01	S01	2	1,599.95
ORD100002	2024-01-02	P03	S01	1	116.66
ORD100003	2024-01-02	P08	S08	2	114.47
ORD100004	2024-01-03	P08	S05	3	179.71
ORD100005	2024-01-03	P11	S03	1	13.68
ORD100006	2024-01-04	P05	S02	1	319.89

Product table

ProductID	Product Name	Product Category	Unit Price
P01	Laptop	Electronics	850
P02	Smartphone	Electronics	600
P03	Headphones	Electronics	120
P04	Office Chair	Furniture	210
P05	Desk	Furniture	320
P06	Bookshelf	Furniture	150
P07	T-Shirt	Apparel	25
P08	Jeans	Apparel	60
P09	Jacket	Apparel	140
P10	Coffee Beans	Grocery	18
P11	Olive Oil	Grocery	14
P12	Snack Box	Grocery	22

Store table

StoreID	Store Name	Region	City
S01	RetailPulse Downtown	North	Chicago
S02	RetailPulse Lakeside	North	Minneapolis
S03	RetailPulse Riverside	South	Houston
S04	RetailPulse Sunbelt	South	Atlanta
S05	RetailPulse Harbor	East	Boston
S06	RetailPulse Metro	East	New York
S07	RetailPulse Pacific	West	Seattle
S08	RetailPulse Canyon	West	Phoenix

Practical 7: Import and Transform RetailPulse Data in Power BI

Objective: Explore the Power BI Desktop interface, import the RetailPulse workbook, and clean and shape it using Power Query.

Operations performed in Power BI Desktop

1. Opened Power BI Desktop and identified the **Report, Table and Model views**, the ribbon, and the **Data, Visualizations and Filters** panes.

2. Home > Get data > Excel workbook; opened RetailPulse_Sales.xlsx, ticked Sales, Product and Store in the Navigator and clicked **Transform Data**.

3. In Power Query Editor set data types: **Sales Date = Date, Quantity = Whole Number, Sales Amount = Fixed decimal number**.

4. Home > Remove Rows > Remove Blank Rows and **Remove Duplicates** on Sales (blank rows found: 0; duplicate rows found: 0); **Replace Values** null -> 0 in Quantity (nulls found: 0). Column headings were already clear.

5. Add Column > Conditional Column named **Sale Size**: if Sales Amount >= 1000 then "Large" else "Standard".

6. Reviewed **Applied Steps**, clicked **Close & Apply** and saved as **RetailPulse_Analysis.pbix**.

Applied Steps (Sales query)

#	Applied step	Purpose
1	Source	Connect to Excel workbook
2	Navigation	Select the Sales table
3	Changed Type	Date / Whole Number / Fixed decimal
4	Removed Blank Rows	Drop fully empty rows
5	Removed Duplicates	Drop repeated rows
6	Replaced Value	Null Quantity -> 0
7	Added Conditional Column	Sale Size (Large / Standard)

Data types after transformation

Table	Column	Type
Sales	Sales Date	Date
Sales	Quantity	Whole Number
Sales	Sales Amount	Fixed decimal number
Sales	OrderID / ProductID / StoreID	Text
Sales	Sale Size	Text
Product	Unit Price	Fixed decimal number
Store	Region, City, Store Name	Text

Outcome: All three tables loaded without errors; dates are recognised as dates and amounts as numbers; Sale Size labelled 240 rows as Large; the dataset was already clean, so no blank or duplicate rows were removed (the steps are still recorded and will re-run on refresh).

Practical 8: Model, Calculate, and Publish a RetailPulse Sales Report

Objective: Create relationships, write basic DAX, build interactive visuals, and publish the report.

Operations performed

1. Model view: created one-to-many relationships Product[ProductID] -> Sales[ProductID] and Store[StoreID] -> Sales[StoreID], single-direction filtering.

2. Home > New measure for Total Sales, Total Quantity and Average Order Value; **New column** for Sales per Unit.

3. Built the report page: card (Total Sales), bar chart (Sales by Product Category), line chart (Sales by month), Region slicer.

4. Tested cross-filtering by clicking a bar and changing the slicer; added titles and axis labels.

5. Home > Publish, selected a workspace and shared the report with the instructor (subject to workspace permissions).

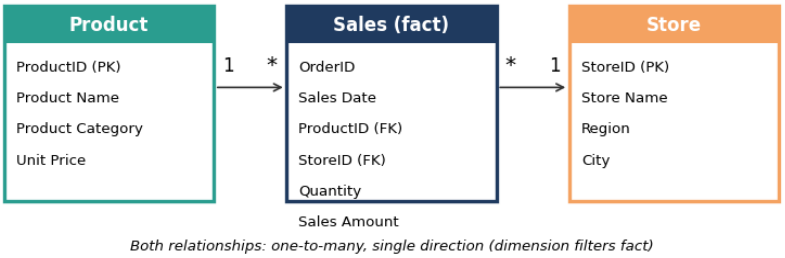


Figure 1: Model view of the RetailPulse star schema.

DAX written

```
Total Sales = SUM(Sales[Sales Amount])
Total Quantity = SUM(Sales[Quantity])
Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))
Sales per Unit = DIVIDE(Sales[Sales Amount], Sales[Quantity]) (calculated column)
```

Measure results (Region = All)

Measure	Value
Total Sales	$812,474.31
Total Quantity	5,414
Average Order Value	$451.37

Practical 8: Dashboard Output

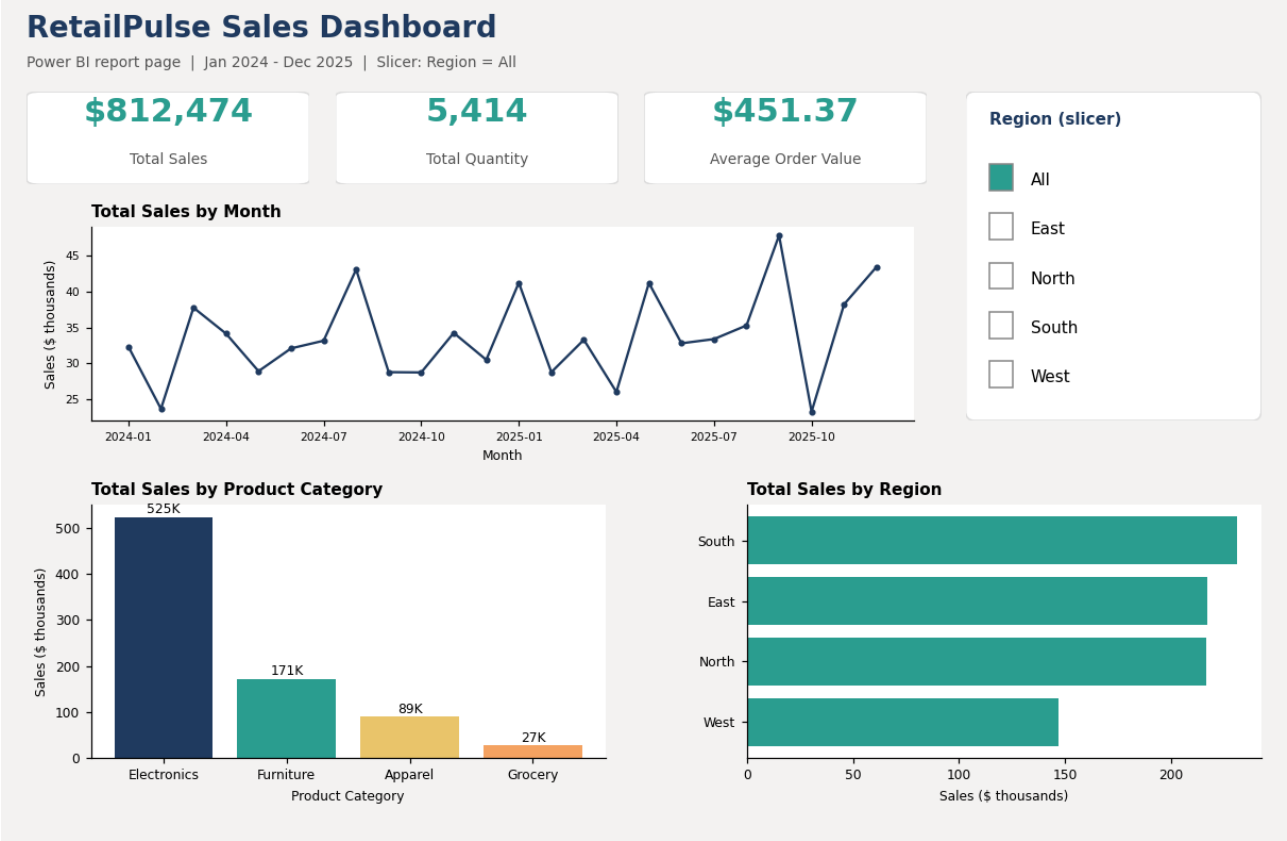


Figure 2: RetailPulse dashboard page with card visuals, category bar chart, monthly line chart and Region slicer.

Total Sales by Product Category (sum equals the card value)

Product Category	Total Sales
Electronics	$524,579.39
Furniture	$171,362.23
Apparel	$89,425.66
Grocery	$27,107.03
Total	$812,474.31

Outcome: Relationship markers (1 and *) sit on the correct ends; the Total Sales card equals the sum of the category bars; measures respond to the Region slicer; the report is published to the chosen workspace and visible to others only when given access.

Practical 9: Time Intelligence and Advanced DAX for RetailPulse

Objective: Create a date table, time intelligence and context-modifying measures, and use Performance Analyzer.

Operations performed

1. Modeling > New table: created the Date table, added Year, Month Number and Month Name columns, marked it as a date table and related Date[Date] to Sales[Sales Date].

2. Created **Sales YTD** and **Sales LY** (TOTALYTD, SAMEPERIODLASTYEAR).

3. Created **YoY Growth %** using VAR / RETURN.

4. Created **Region Share %** using CALCULATE with ALL.

5. Built a matrix: Year and Month on rows; Total Sales, Sales YTD, Sales LY and YoY Growth % as values.

6. Opened **View > Performance Analyzer** and recorded visual load times (see section below).

DAX written

```
Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date]))
Year = YEAR('Date'[Date])
Month Number = MONTH('Date'[Date])
Month Name = FORMAT('Date'[Date], "mmm")

Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth % = VAR Curr = [Total Sales] VAR Prev = [Sales LY]
  RETURN DIVIDE(Curr - Prev, Prev)
Region Share % = DIVIDE([Total Sales],
  CALCULATE([Total Sales], ALL(Store[Region])))
```

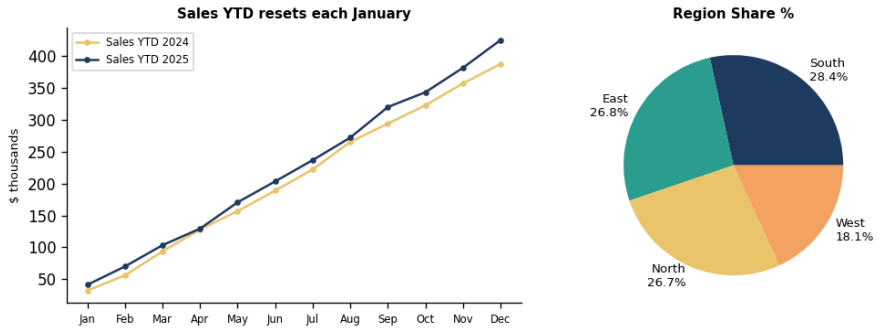


Figure 3: Sales YTD by year and Region Share %.

Region Share % (totals 100%)

Region	Total Sales	Region Share %
South	$231,140.76	28.4%
East	$217,416.25	26.8%
North	$216,650.81	26.7%
West	$147,266.49	18.1%
Total	$812,474.31	100.0%

Practical 9: Matrix Output

Year	Month	Total Sales	Sales YTD	Sales LY	YoY Growth %
2024	Jan	32,326	32,326		
2024	Feb	23,661	55,987		
2024	Mar	37,784	93,771		
2024	Apr	34,159	127,930		
2024	May	28,915	156,845		
2024	Jun	32,099	188,944		
2024	Jul	33,169	222,113		
2024	Aug	43,112	265,225		
2024	Sep	28,778	294,003		
2024	Oct	28,742	322,745		
2024	Nov	34,261	357,007		
2024	Dec	30,504	387,511		
2025	Jan	41,260	41,260	32,326	27.6%
2025	Feb	28,779	70,039	23,661	21.6%
2025	Mar	33,301	103,339	37,784	-11.9%
2025	Apr	26,016	129,355	34,159	-23.8%
2025	May	41,227	170,582	28,915	42.6%
2025	Jun	32,808	203,391	32,099	2.2%
2025	Jul	33,396	236,787	33,169	0.7%
2025	Aug	35,301	272,088	43,112	-18.1%
2025	Sep	47,867	319,955	28,778	66.3%
2025	Oct	23,261	343,217	28,742	-19.1%
2025	Nov	38,252	381,469	34,261	11.6%
2025	Dec	43,495	424,963	30,504	42.6%

2024 rows show blank Sales LY and YoY Growth % because no 2023 data exists. Sales YTD restarts in January 2025.

Performance Analyzer

Recording steps: View > Performance Analyzer > Start recording > Refresh visuals > note each visual's duration, then optimise and record again. Optimisations applied to the model: remove unused columns, reduce the number of visuals per page, prefer measures over calculated columns, and use VAR in measures.

Timings are machine-specific, so record them on your own Power BI Desktop and fill in below:

Visual	Before (ms)	After (ms)
Matrix		
Card		
Bar chart		
Line chart		

Outcome: Sales YTD accumulates within each year and resets each January; Sales LY equals the previous year's Total Sales for the same period; Region Share % totals 100%.